

?Chapter Title: CAN ENGINEERING ETHICS BE TAUGHT

Book Title: Engineering Ethics

Book Subtitle: Contemporary and Enduring Debates

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Published by: Yale University Press

Stable URL: <https://www.jstor.org/stable/j.ctv10sm953.5>

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PART ONE

Foundational Issues

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1 CAN ENGINEERING ETHICS BE TAUGHT?

IN 2010, after a two-year inquiry, a judge concluded that Canadian prime minister Brian Mulroney had acted inappropriately when he accepted large amounts of cash from a German Canadian arms lobbyist. The judge suggested that all public servants should get ethics training. Peter Worthington, a columnist for the *Toronto Sun*, responded to this suggestion in the following way: “A case can be made that ‘ethics’ are something that you either have, or you don’t have. Or, to put it slightly differently, ethics are a code you subscribe to or chose to ignore for reasons of personal interest. . . . All the training, teaching, studying, reading, or lectures in ‘ethics’ will not make a person more ethical if he or she does not have these core values to begin with.”¹

Reading between the lines, we might think that Worthington believes people acquire their core moral values during their childhood. Once they reach a certain age, not much can be done. If a person was brought up to have ethics, great; if, on the other hand, someone didn’t have that kind of upbringing, forget it—the person will never change and never learn to become ethical.

Worthington expresses a form of skepticism that is not uncommon when it comes to teaching ethics to undergraduate engineering students. For example, Karl Stephan, a professor of engineering, described the following encounter: “Some years

ago I argued with a fellow professor about the issue of engineering ethics education at the college level. His point was along the lines of, 'Hell, if eighteen-year-old kids don't know right from wrong by the time we get 'em, they're not going to learn it from us.'"²

Despite such skepticism, many (if not most) undergraduate engineering programs in the United States, as well as in many other countries, require training in ethics as part of the curriculum. One reason they do so, is that ABET, the accreditation organization for undergraduate engineering programs in the United States, requires it.³ ABET specifies a list of outcomes that students must achieve by the time they graduate. These include "an understanding of professional and ethical responsibility" and acquiring "the broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context." Programs demonstrate their achievement of the outcome in a variety of ways, including semester-long courses on engineering ethics.

If the skeptics are right, ABET's requirements about ethics are a waste of time. If ethics cannot be taught, there is no point in requiring that engineering programs teach it. So, the question whether ethics can be taught is important for engineering as well as more broadly.

The question is not a simple one, at least not as simple as the skeptics suggest, and in exploring it in this chapter, we will necessarily go deeper into the goals of ethics education. We can, for a start, dismiss the idea that the goal of ethics education—in engineering as well as other fields—is to ensure that no one—not one single engineer—will ever do anything wrong. Skepticism is appropriate for this impossible goal. Yet there are more modest goals that, if achieved, would increase the likelihood of individuals behaving ethically. For example, in engineering one goal might be to inform engineering students about the codes of ethics promulgated by engineering professional societies. Ensuring that students know about these codes and that they are considered important by engineering professional organizations increases the likelihood that students will follow them.

Some readers may be surprised that the first debate in a book on engineering ethics is whether ethics can be taught. Indeed, some may think that this chapter is only for ethics instructors. However, the chapter is placed first for a purpose: it sets expectations for what is to follow. Sorting out what can and can't be taught and what are and are not appropriate goals for ethics education prepares readers for what will follow in subsequent chapters.

MOTIVATIONS FOR MORAL BEHAVIOR

Although this chapter focuses on the question whether engineering ethics can be taught, we begin with a debate about what motivates moral behavior. At the core of the question about whether ethics can be taught is a set of beliefs about the sources or causes of moral behavior. In other words, in order to answer the question about what can be taught, we must ask a prior question about what goes on in a person when he or she makes a moral decision. Skeptics seem to eschew this question by insisting that what motivates moral behavior is something in individuals that is fixed and not amenable to influence—either ever or after a person has reached a certain age. By contrast, ethics education presumes that ethical behavior can be influenced, and it presumes that the target—the thing to be influenced—is how people make moral decisions. Ethics education can make people better, the presumption is, by improving how they make moral decisions. Many believe that reasoning is a significant part of this; hence, ethics courses are often designed to improve ethical reasoning. Curricula and courses aim to provide students with information, ideas, and experiences that will improve their ethical reasoning skills. In fact, many ethics courses focus on ethical theories such as utilitarianism, deontology, and virtue ethics precisely because these theories are thought to be tools and heuristics for improved ethical reasoning. (These theories will be discussed in chapter 3.)

Recently, however, the focus on ethical reasoning has been challenged. Critics claim that ethical behavior is based largely on moral intuitions, not reasoning. Moral intuitionists (also referred to as social intuitionists because they believe that moral intuitions

are interpersonal) claim that if we look at how people make moral decisions, we will find that intuitions play a much larger role than previously thought. Their position is aligned with skeptics insofar as it suggests that teaching moral reasoning will have little or no effect. Moral intuitionists argue that reasoning comes in only after intuitions; reasoning is, they say, more like rationalization in the sense that it is used to explain and justify what one's intuitions tell one to do. Jonathan Haidt put it succinctly when he wrote, "The central claim of the social intuitionist model is that moral judgment is caused by quick moral intuitions and is followed (when needed) by slow, *ex post facto* moral reasoning."⁴

Moral intuitionists often point to human experience as evidence for their position; they point to individuals who are repulsed by the sight of certain kinds of behavior. For example, they claim that without reasoning individuals find the idea of incest or torture of children abhorrent, and that for some the thought of eating animals is viscerally distasteful. These experiences suggest that a person's ethics is based on her or his intuition, not reasoning. This view is in line with skepticism about teaching ethics in the sense that it claims that ethical behavior is the result of something somewhat primitive in humans, something not amenable to reason.

Nevertheless, there are some serious criticisms of moral intuitionism, criticisms that take us deeper into the matter of what motivates moral behavior. One counter is that moral intuitions don't come out of nowhere; they aren't inborn. Reasoning comes into play as individuals form their moral intuitions. According to this line of thinking, individuals acquire their intuitions through developmental processes that involve reason. In other words, one's experiences and reasoning about those experiences shapes one's intuitions. If this is right, then addressing moral reasoning in education might lead to better moral intuitions.

Another line of criticism of moral intuitionism points to some obvious cases in which reasoning seems to change a person's moral intuitions. For example, through discussion and debate with a friend, one might be convinced to become a vegetarian, or one might, through reflection, decide to refrain from telling certain kinds of jokes, say, because they perpetuate racial or gender ste-

reotypes. The point here is that whatever role intuition may play in ethical decision-making, it can be tempered by reasoning.

Yet another criticism points to situations that are not so simple (and arguably more common) in which the right thing to do is more complex and challenging. As David Pizarro and Paul Bloom explain, Haidt “was likely correct that we do have quick and automatic responses to certain situations—killing babies, sex with chickens, and so on—and although these responses can be modified and overridden by conscious deliberation, they need not be. But most moral cognition is not about such simple cases; in the real world, moral reasoning is essential.”⁵

In these more complicated situations, intuitions may not help. Imagine having to decide whether to blow the whistle on your boss for apparently illegal behavior or deciding whether to do what a friend has asked even though you think the friend is, at the moment, not mentally stable. In such situations, individuals are compelled to think carefully about the details and the consequences of taking this or that line of action. In these circumstances, being able to think through the situation, reason about alternative options, and anticipate consequences is not just helpful but essential.

Although Pizarro and Bloom’s point seems correct, as do the other criticisms of moral intuitionism, moral intuitionists seem right to insist that moral behavior involves more than reason alone. Unfortunately, the debate cannot be resolved here. Indeed, the dialogue between moral intuitionists and moral reasoning advocates is a recent manifestation of a philosophical dialogue that has been going on for several thousand years. In earlier manifestations, the question was whether reason is a slave to the passions or vice versa. The point of exploring this debate is to show that the question whether and how ethical behavior can be influenced is a deep and enduring matter and not so easily dismissed as the skeptics would have it.

We turn now to another way of answering the question. We can look at what can and is being done to teach ethics. After considering what ethicists take to be the goals of teaching engineering ethics, we will be in a better position to determine whether education can influence moral behavior.

CAN ENGINEERING ETHICS BE TAUGHT?

We can begin by considering the idea that engineering ethics may have a knowledge component. That is, setting aside both reasoning and intuition, consider that being an ethical engineer may require information. If so, then there is at least one aspect that can be transmitted by means of education—that is, this one component can be taught.

We know that engineering requires mastery of a body of knowledge. That is, to do much of the work that engineers do, one needs technical knowledge and know-how. The knowledge or know-how needed depends on the nature of the project and the field of engineering, but engineering—at least modern engineering—is an activity that requires a good deal of expert knowledge. Although this is not often conceived as part of engineering ethics, in fact competence in engineering is essential to ethical engineering. For example, one can't build a safe bridge, identify environmental pollution, or design life-saving biomedical devices unless one has the technical expertise. In fact, many codes of ethics of engineering professional organizations specify that engineers should maintain and improve their technical expertise and should only perform jobs in areas of their competence. This is important to note: competence in engineering is part of ethical engineering, so technical knowledge is part of engineering ethics.

Nevertheless, technical knowledge and expertise do not alone or necessarily lead to ethical engineering. Engineering ethics is directed at how technical competence is deployed and at understanding the contexts and relationships in which it is used. This requires knowledge, but of a different sort. For a start, the knowledge component of engineering ethics involves knowledge of professional norms and practices; knowledge of the expectations of employers, clients, and the public; knowledge of how to manage these relationships; and knowledge of the social consequences of technical endeavors.

To better understand how knowledge is relevant to engineering ethics, we can consider a case from the National Society of Professional Engineers' (NSPE) repository of cases. The NSPE

has constituted a Board of Ethical Review (BER) consisting of seven licensed members appointed by the society's president, and each year the board reviews cases that have been submitted. Whether the cases are real or hypothetical is not known. The BER analyzes each case and states whether the behavior described in the case is ethical or unethical. The analysis is meant in part to interpret the NSPE Code of Ethics. (Engineering codes of ethics will be detailed in chapter 2.)

Consider NSPE Case 15-10:

Engineer A, a professional engineer, works as the director of the local government building department. Engineer A also has a part-time sole engineering practice and prepares a set of structural engineering drawings for Client X. The drawings must be approved by the local building department. Engineer A does not participate in the review or approval of the drawings but Engineer A's assistant, Engineer B, a professional engineer, reviews and approves the engineering drawings prepared by Engineer A.⁶

In reviewing this case, the BER concludes that both Engineer A and Engineer B behaved unethically.

What exactly did the engineers do wrong and why? Engineer A seems vaguely aware of the concept of conflict of interest and entirely unaware of how to handle such situations. A conflict of interest is when a person has a role that requires the exercise of judgment on behalf of others but also has an interest that might taint, or merely appear to taint, his or her judgment. (Conflicts of interest will be discussed in more detail in chapter 4.) In this case, Engineer A is expected to act as a representative of the local government building department—that is, he is expected to act in that department's interests. However, he has a personal interest in having the drawings approved, and that personal interest has the potential to influence his judgment. Hence, he should not be the one who reviews the drawings. The objectivity of his judgment could easily be called into question.

Engineer A seemed to understand that there was a problem in his reviewing his own plans because he asked his assistant to do the review. However, in doing this, Engineer A made things worse in two ways. First, he didn't inform his employer about the

conflict of interest. The BER describes Engineer A's behavior as a failure to "fulfill his obligation to act as a faithful agent or trustee to his employer." Second, Engineer A compounded the problem by asking someone he supervised to review the drawings. The BER is unambiguous in specifying that Engineer A should not have had Engineer B review the drawings because Engineer A supervised Engineer B. The BER explains, "It would be unethical for Engineer A to provide the services in the manner indicated, even if he had obtained approval from his supervisor, because he cannot require his subordinates to approve his work." The BER insists that a subordinate is not capable of rendering objective (untainted) judgment on a supervisor's work.

This case suggests the importance of knowledge to engineering ethics in that Engineer A might have behaved differently had he known about the NSPE Code of Ethics, what constituted a conflict of interest, and what the norms of professional practice are for someone in his position. Of course, it is possible that Engineer A knew all of this and still did what he did. In this respect, knowledge does not make a difference to someone who is willfully unethical. Nevertheless, for those who wish to do the right thing, knowledge is crucial. Individuals are less likely to behave unethically if they are aware of their ethical obligations—that is, the norms and practices in their field.

The same could be said for Engineer B. She might have behaved differently had she known about norms of professional practice. Again, we can't be sure whether Engineer B was merely ignorant of professional norms or knew the norms but felt pressured to violate them because of her supervisor's request. If it was ignorance, then knowledge of professional norms and training in how to deal with difficult situations might have helped her respond to Engineer A's request. If, on the other hand, she knew she shouldn't do the review but felt pressured, she could have pointed to the NSPE Code of Ethics and would have been better equipped to articulate why she had to refuse Engineer A's request.

This case illustrates that knowledge (not just technical knowledge) has a role to play in engineering ethics. It can make a difference in how people behave. In this case, knowledge of engineering

professional codes and norms of professional practice might have affected the behavior of both engineers. In other cases, other kinds of knowledge might be relevant to an engineer's decision. To be sure, as already indicated, perhaps both engineers had the knowledge and chose to ignore it. For them, something more than knowledge would be needed to discourage their unethical behavior.

Next we'll delve more deeply into the knowledge component of engineering ethics and see that engineering ethics education can do more than provide knowledge.

THE GOALS OF TEACHING ENGINEERING ETHICS

Within the burgeoning literature on engineering ethics education, as well as professional ethics education more broadly, there is much debate about what can and should be taught and how best to teach it. For example, some think that the primary goal should be to make students aware of the ethical issues that might arise in professional practice. Others think that the goal should be to improve ethical decision-making and judgment. Yet others think that the emphasis should be on motivating and even inspiring students to behave ethically. Of course, these goals are not mutually exclusive. Four types of goals come up persistently in the literature. Engineering ethics education should: (1) provide knowledge of codes and standards; (2) increase awareness of ethical issues and develop the ability to identify ethical issues; (3) provide training in ethical decision-making; and (4) inspire the motivation to be ethical.

Of course, skeptics might still insist that none of these goals is achievable because education can't counter poor moral upbringing or bad character. For that reason, it is important to examine each goal more fully and consider the effect that each aims to achieve.

Knowledge of Codes of Professional Conduct

The discussion of NSPE Case 15-10, above, illustrates the first goal. This case shows that knowledge of a professional code of ethics and, more broadly, norms of professional practice might

make a difference in how engineers behave. However, more than an awareness of codes of ethics and standards of behavior may be needed. The codes of ethics and standards of behavior must be understood. They must be grasped in a way that allows them to be used in or applied to particular situations. In other words, although familiarity with codes of ethics is essential in preparing for the ethical challenges that engineers face as professionals, exposure and familiarity are not enough. Engineering ethics education should provide training in how to interpret and apply the statements in the codes.

To better understand the importance of learning to interpret and apply standards and codes of ethics, consider a criticism put forward by Stefan Eriksson, Gert Helgesson, and Anna Höglund.⁷ These authors focus on health care ethics, but their critique applies to all professional ethics. Their concern is with an overreliance on codes of ethics combined with insufficient attention to the problem of interpreting them. Simply knowing what a code of ethics says doesn't help people in real-world situations, because in order to see how a code or rule applies to a situation, the code or rule must be interpreted. For example, in Case 15-10, Engineer A may have known that the NSPE Code of Ethics specifies that engineers are to act as faithful agents of their employers, but he may not have made the connection between having his drawings approved and being faithful to his employer. He may not have thought about how his employer's interest might be in tension with the interests of his private practice clients. Simply put, he might not have really understood the edict to be faithful to one's employer.

Consider a different situation that can arise in engineering practice. Many engineering codes of ethics specify that engineers should act faithfully on behalf of their clients. Imagine an engineer who discovers a safety problem while working on a client's building. The safety problem could affect people who work in the building, say, by increasing the risk of fire. The engineer tells her client and the client asks the engineer to keep quiet about the safety issue because the client is planning to sell the building. Although the codes of ethics make it clear that engineers have a responsibility to their clients, the codes also specify that engineers

should hold paramount the safety, health, and welfare of the public. In this situation, simply knowing that a code of ethics exists and knowing what it says is not enough to help the engineer figure out what to do. She may well know of the commitment to hold the safety of the public paramount as well as the commitment to be faithful to clients and still not know how to handle a situation when two professional norms seem to conflict.

Of course, there may be no simple answer to the engineer's dilemma, but familiarity with similar situations and practice thinking through those situations might help the engineer. Charles Abaté argues that students should be required to engage in activities that involve interpreting and debating the meaning of the codes so that they will have experience in how to apply them.⁸ Moreover, in practicing how to interpret and apply particular norms, students learn strategies that can be used in similar situations. Engineering education should, therefore, give students opportunities to grapple with the meaning and implications of statements in codes of ethics and practice interpreting and applying them.

Another problem with knowing only that codes exist and not what they mean and how they apply is that the lack of understanding may lead to legalistic thinking. This is a problem both because law and ethics are not the same and because legalistic thinking may suggest that one should simply follow the rules without consideration of aspects of a situation that aren't covered. Codes of ethics are intended to provide guidelines and not to be rules that one must follow no matter what.

The criticism of teaching codes of ethics made by Eriksson, Helgesson, and Höglund is not, then, a rejection of the goal of raising awareness of codes of ethics; rather, it is an argument for doing this in a way that develops skill at interpreting and applying the codes and doesn't lead to an overemphasis on legalistic thinking.

Awareness and Ability to Identify Ethical Issues

Engineering codes of ethics are critical to engineering and to engineering ethics, but not all ethical issues can be covered in a code. Another important goal of engineering ethics education

is to cultivate the ability to identify ethical issues. Consider the following case.⁹

Joe is employed as a manufacturing engineer. In this role, he meets periodically with vendors who supply his company with materials. It is common to chat informally with vendor representatives, and Joe discovers that one of the representatives—Frank—is an avid golfer, as is Joe. During one of their meetings, Frank invites Joe to play golf at a posh country club that Joe knows has an excellent golf course. Joe takes Frank up on the offer, and they have a competitive but friendly match with a few other players as well. Frank invites him again and suggests that they play for small amounts of money. Although Joe is reluctant, he agrees. The practice of playing golf together continues; Joe is surprised, but he often wins money on the games. After some time, Frank nominates Joe for membership in the country club, and his membership application is approved.

This continues for several years, and then Joe is called into a meeting and told that his company is having financial difficulty and needs to institute cost-cutting measures, including termination of several vendor contracts. Joe is assigned to work with two other engineers to decide which vendors will be cut. Because of his relationship with Frank, Joe is now worried about his objectivity in making this decision. He tells the other two engineers about his relationship with Frank and continues to work with them. Joe concludes that it would be best to eliminate Frank's company; the other engineers independently arrive at the same decision.

Joe decides that he must tell Frank what has happened, so he meets with Frank and tells him the news. Frank is quite upset and wants to know what Joe did to try to protect Frank's company. Frank becomes angry and blurts out: "I don't believe this! What kind of friend are you, anyway? Didn't I get you into the country club? And how good a golfer do you think you are, anyway? How do you think you've won all that money from me over the years? You don't really think you're that much better at golf than I am, do you?"

In this case, Joe was unaware of the pitfalls of establishing a personal relationship with a vendor. He didn't recognize that entering into a relationship outside work might take him into delicate

ethical territory. He didn't think to question Frank's motives for increasingly involving him in a golfing relationship, nor did it occur to him that he might someday have to make a tough professional decision that would affect Frank. This is a case in which the engineer might have benefited from having explored the ethical issues that can arise in professional relationships. Joe might have picked up on Frank's motives sooner or anticipated potential problems had he been more aware of the nuances of professional relationships. The case is intriguing because it is difficult to determine when Joe went too far, if ever. Should he have recognized a potential problem sooner? Should he have said no to the invitation to play golf? To play for money? To join the golf club?

Analysis of case studies like this can raise awareness of ethical issues and sharpen one's ability to identify ethical issues before they become difficult and complicated. Exposure to case studies—whether real or fictitious—increases the likelihood that a new engineer will identify ethical issues quickly and early on before they cause too much trouble. Abaté argues that analysis of case studies is effective because it draws on a basic human capacity for pattern recognition.¹⁰ Exposure to case studies allows one to discern patterns that can then be seen in other, real-world situations.

Another strategy that helps to identify ethical issues is familiarity with moral concepts and frameworks. The idea here is that when one has a vocabulary for thinking about ethics, one is more likely to recognize situations involving ethics. Thinking carefully about moral concepts such as integrity, loyalty, respect, or conflict of interest and their implications increases the likelihood that one will see the relevance of these concepts to real-world situations. For example, in chapter 3, when we discuss ethical concepts and theories in more detail, we will explore the concept of never treating a person merely as a means. One is much more likely to notice when someone is being treated in that way when one has discussed this concept and has had experience in using it. Ethical concepts and theories provide language, principles, and ways of looking at the world that help people notice what they might otherwise ignore.

Moral Reasoning and Improved Decision-Making

The ability to identify ethical issues is an important part of being an ethical engineer. However, it is one thing to see that there is an issue and quite another to know what to do about it. A central focus of engineering ethics is on reasoning and ethical decision-making. This takes us back to the skeptics, moral intuitionists, and questions about what is going on inside someone when he or she makes a moral decision.

Without a doubt, engineers often face situations in which they must make tough moral decisions: Should I blow the whistle on my employer? Should I violate confidentiality when my client appears to be engaged in illegality? What should I do if asked to pay to play in order to get a contract? Should I let this design go to the next level of development when I believe that the design will never be safe enough to be built? So, the question isn't whether engineers engage in moral decision-making. They do! The question is whether and how ethical decision-making can be taught.

Since engineering ethics education is relatively new, there is still some uncertainty about how best to teach it, and in this context, ethical decision-making poses the greatest challenge.¹¹ The standard approach is to provide practice—practice with feedback and discussion. Here again case studies and case analysis are considered the best approach. Courses that provide the opportunity to consider, reflect upon, analyze, discuss, and even debate what to do in tough situations are thought to improve decision-making skills. Engineering ethics courses often include case or scenario analysis and role-playing, providing students an opportunity to experience ethical dilemmas and reflect on how to handle these situations. The idea is that calm and careful reflection on situations and hearing the ideas of others will prepare engineers for the real world in which there may be pressure to act quickly and fewer opportunities for discussion and reflection.

Motivation

Another goal of engineering ethics is to inspire and motivate ethical behavior. The argument for this component is straightforward.

ward. Providing people with knowledge, skills, and experience isn't enough to ensure that they will act on the knowledge or exercise the skills. They must be motivated to be willing to use their knowledge and skills especially in challenging situations, circumstances that may require acting against their self-interest.

Little is formally known about how best to inspire and motivate. Yes, we all have had the experience of being inspired by a charismatic leader, a passionate speech, or a life-changing event, but how can this be translated into engineering education? Some ethicists have suggested that it might be done by exposure to exemplary and heroic engineers.¹² The idea here is that hearing about engineers who have made great sacrifices to do the right thing or who have taken on noble causes or who have withstood pressure to engage in wrongdoing will inspire other engineers to do likewise. William LeMessurier is often put forward as such an inspiring figure. LeMessurier was a distinguished structural engineer and a consultant in the development of the innovative Citicorp Headquarters building in New York. After the building was completed (in 1977), LeMessurier learned that it did not meet the safety standards for buildings situated as it was. As Michael Pritchard explains, LeMessurier "knew how to correct the problem, but only at the cost of a million dollars and at the risk of his career if he were to tell others about the problem. He promptly notified lawyers, insurers, the chief architect of the building, and Citicorp executives. Corrections were made, all parties were cooperative, and LeMessurier's career was not adversely affected."¹³

LeMessurier is inspiring because he took on personal risk in order to ensure the safety of a building that he had designed. Several engineering ethicists have compiled stories of heroic and exemplary engineers who are less well known than LeMessurier but equally impressive in how they risked their own well-being in order to protect others.¹⁴

Inspiration is related to moral courage. Engineers often find themselves in situations in which their work environment makes it difficult for them to express their ethical concerns: a concern about safety, an awareness that vulnerable populations will be harmed by a project, perhaps a suspicion that something illegal

is going on. Moral courage is the ability to act for moral reasons and despite one's fear or the likelihood of personal harm. Moral courage is the capacity to speak out and to stand up for what one thinks is the right thing to do. It may well be that most people want to do the right thing but not all people have the moral courage to act in tough situations. Inspiration may well promote moral courage.

CONCLUSION

Can engineering ethics be taught? The skeptics who answer this question in the negative seem to have an overly simplistic picture of human behavior. To suppose that ethics can only be taught in childhood and that nothing after that can affect a person's moral thinking or moral behavior seems not to acknowledge or appreciate how people engage in ethical decisions. Moreover, it seems misguided to suppose that ethics is simply a matter of learning a few rules in early childhood and then following them. To be sure, learning a few basic moral principles early on in life is probably a good thing, but real-world situations are often complex and require more than following general rules. Rules need to be interpreted, and it is not always easy to see which rule applies when and how.

In this chapter we have seen that engineering ethics has an informational or knowledge component (for example, know what the norms for engineering are); a cognitive component (for example, recognize when there is an ethical issue); a reasoning component (for example, make a moral judgment); and a motivational component (for example, move yourself to act). All four dimensions can be addressed through education.

Engineering ethics can be taught by providing knowledge of professional codes and how to interpret and apply these codes to real-world situations. Awareness of ethical issues can be heightened, as can the ability to identify ethical issues. Training and experience in ethical decision-making can be provided. And engineers can be inspired to want to be ethical and to act in morally courageous ways. Although all of this may not make so-called bad people good or ensure that all engineers will always do the

right thing, pursuit of these activities increases the likelihood that engineers will be better able to handle the ethical issues that arise in their professional lives.

This chapter has focused on individuals, specifically, on whether education can influence the behavior of individuals. However, engineering ethics is not just a matter of individual behavior. It is also a matter of institutions, organizations, and environments enabling individuals to behave in certain ways. Organizations and institutions can do a lot to promote and facilitate ethical engineering. In the next chapter we turn our attention to engineering as a profession and consider the activities of engineering professional organizations and codes of ethics in promoting and facilitating ethical engineering.

SUGGESTED FURTHER READING

Articles in *Science and Engineering Ethics*

Articles in *Journal of Professional Issues in Engineering Education and Practice*

Barry, Brock E., and Joseph R. Herkert. "Engineering Ethics." Pages 672–92 in *Cambridge Handbook of Engineering Education Research*. Cambridge: Cambridge University Press, 2015. <https://doi.org/10.1017/CBO9781139013451.041>.

National Society of Professional Engineers, BER Cases, <http://www.nspe.org/resources/ethics/ethics-resources/board-of-ethical-review-cases>